

Reducing Hallucination in Retrieval-Augmented Generation Using Hybrid Vector–Graph Integration with Evidence Grounding

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Abstract - *Large Language Models (LLMs) are widely used for text generation and analysis, but they often produce hallucinated responses that are not supported by factual evidence. Retrieval-Augmented Generation (RAG) improves reliability by grounding responses in external data; however, most systems rely on a single retrieval method, which limits their ability to capture related information distributed across documents. This paper explores whether combining semantic vector retrieval with graph-based retrieval can improve evidence grounding and reduce unsupported responses in document-based question answering systems. A hybrid retrieval framework is designed in which vector similarity captures semantic relevance, while graph traversal captures structural relationships between entities. The system incorporates citation mechanisms to link responses to source evidence and abstention mechanisms to avoid answering when sufficient evidence is not available. The approach is evaluated using a structured query set that includes factual, multi-hop, and unanswerable cases. Evaluation metrics measure retrieval quality, grounding accuracy, and hallucination behavior. The results demonstrate that hybrid retrieval achieves equivalent answer quality to a vector-only baseline while actively contributing graph-sourced evidence in 31.4% of query responses, confirming that knowledge graph integration is additive rather than substitutive. The system maintains Recall@10 at 95.3%, Abstention Accuracy at 87.3%, and Unsupported Claim Rate at 12.7% across both configurations. These findings indicate that integrating semantic and structural retrieval signals, along with strict grounding constraints, significantly improves the reliability and trustworthiness of document-grounded LLM systems.*

Keywords - Retrieval-Augmented Generation (RAG), Hybrid Retrieval, Knowledge Graphs, Hallucination Reduction, Evidence Grounding, Large Language Models

I. INTRODUCTION

LLMs are increasingly deployed in document-based workflows such as enterprise search, technical support, policy analysis, and decision assistance. These systems are expected to produce responses that are both accurate and grounded in source documents. However, despite their strong generative capabilities, LLMs often produce hallucinated outputs, meaning responses that are fluent but not supported by factual evidence. This limitation becomes critical in document-based question

answering, where incorrect or unsupported answers can lead to poor decisions, reduced trust, and operational risk.

RAG addresses this challenge by grounding model outputs in external document collections instead of relying solely on parametric knowledge. By retrieving relevant text chunks and conditioning generation on them, RAG improves factual consistency and traceability. However, most RAG systems rely on a single retrieval paradigm, typically dense vector similarity search. While effective at capturing semantic relevance, vector retrieval operates at the chunk level and does not explicitly model relationships between entities or connections across document sections. As a result, it may fail to retrieve all necessary evidence, especially for queries that require multi-hop reasoning or combining information distributed across multiple locations.

This paper addresses this limitation by designing and evaluating a hybrid retrieval framework that combines semantic vector retrieval with graph-based retrieval. The vector component captures semantic similarity between queries and document chunks, while the graph component captures structural relationships such as adjacency and entity-level connections. By combining these complementary signals, the system aims to improve evidence coverage and enable more complete, contextually grounded responses. In addition, the system enforces strict grounding constraints through citation mechanisms that link generated answers to source evidence, and abstention mechanisms that prevent answering when sufficient evidence is not available.

The significance of this work lies in improving the reliability and trustworthiness of LLM-based systems operating over document collections. The project aims to minimize unverified claims by explicitly speaking about retrieval limitations and grounding failures and enhance the transparency of formulated answers.

The rest of the paper will be structured in the following manner. Section II surveys prior research in retrieval-augmented generation and mitigating hallucinations. Section III explains the proposed hybrid retrieval system and methodology. Section IV provides experiment set up and results on evaluation between vector-only and hybrid retrieval. Section V ends with significant findings and the way forward.

II. PRIOR WORK

RAG has emerged as a key approach for improving the factual reliability of LLMs by grounding generated responses in external document collections. Prior work can be broadly categorized into three areas: (i) improving retrieval quality, (ii) reducing hallucinations through reliability-focused mechanisms, and (iii) extending retrieval with structured knowledge representations.

Early research primarily focused on improving retrieval effectiveness so that the language model receives more relevant supporting evidence. Sawarkar et al. proposed Blended RAG, which combines dense vector retrieval with sparse query-based methods to improve ranking quality and retrieval accuracy across benchmark datasets [1]. Similarly, hybrid retrieval strategies have been explored to address the limitations of single-retriever systems, where reliance on semantic similarity alone may fail to capture important contextual signals.

To address structural limitations in vector-based retrieval, recent work has incorporated knowledge graphs into RAG pipelines. Sarmah et al. introduced HybridRAG, which integrates vector retrieval with graph-based reasoning to enhance contextual understanding and factual consistency [2]. Knollmeyer et al. further demonstrated that document-level knowledge graphs can improve question answering by explicitly modeling relationships between entities and document segments [7]. These approaches highlight the importance of combining unstructured semantic retrieval with structured relational information to improve evidence coverage.

Another line of research focuses on reducing hallucinations by improving grounding and model reliability. Koç et al. proposed ReRAG, which adjusts retrieval configurations such as chunk size and overlap using feedback signals to better align retrieved evidence with expected answers [3]. Hu et al. introduced LRP4RAG, which applies Layer-Wise Relevance Propagation to analyze how retrieved evidence contributes to generated responses, improving interpretability and enabling hallucination detection [4]. These approaches emphasize the role of retrieval quality in controlling downstream generation behavior.

In addition to retrieval improvements, several works focus on enforcing reliability through abstention and confidence estimation. Yadkori et al. proposed conformal abstention, enabling models to refrain from answering when confidence is low, thereby reducing the risk of hallucinated outputs [5]. Chan et al. demonstrated that prompt-based confidence scoring can improve calibration and response reliability in RAG-based systems without modifying model architecture [6]. These methods address hallucination at the generation stage by introducing mechanisms that limit unsupported responses.

Overall, prior work demonstrates that improving retrieval quality, incorporating structured knowledge, and enforcing grounding constraints are all critical to reducing hallucination in RAG systems. However, many approaches address these aspects in isolation. This work builds on these directions by integrating hybrid vector-graph retrieval with explicit grounding and

evaluation mechanisms within a unified, end-to-end system, enabling controlled comparison between vector-only and hybrid retrieval strategies.

III. METHODOLOGY

The proposed system follows a staged pipeline that converts raw documents into an evidence-grounded question answering system. The methodology is organized into components covering document processing, representation, retrieval, and grounded generation. Each stage was implemented as a deterministic and restart-safe module so that intermediate artifacts remain reusable, inspectable, and traceable. The overall pipeline integrates document ingestion, semantic indexing, graph construction, hybrid retrieval, and grounded generation.

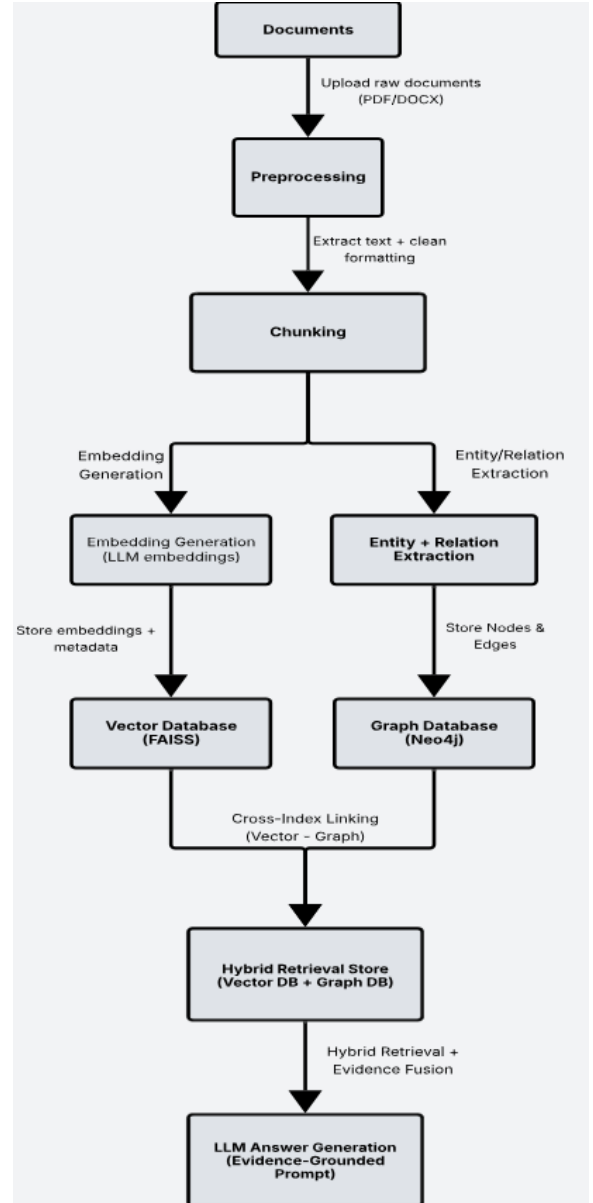


Fig. 1. Hybrid Vector-Graph RAG system architecture
3.1 Document Ingestion and Extraction

As illustrated in Fig. 1, the system processes documents through sequential stages from ingestion to grounded response generation. Document ingestion is treated as a distinct stage to ensure reproducibility and eliminate duplicate processing across repeated runs. Uploaded files are stored in a collection-scoped workspace and assigned deterministic identities using content hashing. This ensures consistent processing and prevents redundant computation.

The extraction stage converts documents into structured text while preserving page-level boundaries required for later citation. PDF extraction combines native parsing with OCR fallback for scanned or complex layouts, while DOCX content is flattened into a structured textual representation. The extracted output is stored as an intermediate processed artifact so downstream stages do not need to re-read the original files.

3.2 Text Cleaning and Normalization

Text normalization is required because raw extracted text contains formatting artifacts that degrade embedding quality and retrieval consistency. Cleaning includes Unicode normalization, whitespace normalization, and removal of control characters while preserving source-faithful text separately for evidence display.

3.3 Semantic Chunking and Representation

Chunking is critical because poor boundaries lead to incomplete or fragmented evidence, directly reducing retrieval accuracy. Documents are segmented using a structure-aware strategy rather than naive fixed-size slicing, ensuring that each chunk represents a coherent semantic unit.

Chunk boundaries follow headings, paragraphs, lists, and tables, with sizes constrained within a predefined range (Table I). Each chunk stores document identity, page number, section path, and character offsets. An `embedding_text` field prefixes content with document and section context to improve retrieval precision. Low-information chunks are filtered using an `index_this` flag to prevent noise from entering the index.

Table 1. System Configuration Summary

Component	Setting
Embedding Model	BAAI/bge-m3
Vector Index	FAISS
Graph Store	Neo4j
Chunk Size	1200–2200 chars
Overlap	450 chars
Reranker	Cross-Encoder MiniLM
Hybrid Weights	$w_v = 0.75, w_g = 0.25$

3.4 Embedding and Vector Indexing

Dense vector representations are used to capture semantic similarity between queries and document content, enabling retrieval beyond exact keyword matching. Each retrieval-eligible chunk is embedded using a transformer-based model and indexed using FAISS for efficient similarity search.

To avoid recomputing identical content, a content-addressed SQLite cache is used, reducing redundant embedding computation across repeated ingestion. Metadata mappings preserve

the linkage between vectors and source spans, ensuring full traceability. The embedding configuration is versioned through an embedder identifier to maintain consistency across pipeline runs (Table I).

3.5 Knowledge Graph Construction

Vector retrieval alone cannot capture relationships across distributed evidence, particularly for multi-hop queries. To address this, a knowledge graph is constructed to model structural and entity-level connections within the corpus.

The graph consists of document, chunk, and entity nodes connected through provenance, adjacency, and semantic relations. Entity and relation extraction is performed using an LLM-based pipeline to adapt to document-specific vocabulary without predefined schemas. Chunk metadata is stamped back onto graph-created nodes to maintain alignment with the vector pipeline. Graph construction is applied only in hybrid retrieval mode to preserve separation from vector-only processing.

3.6 Hybrid Retrieval Mechanism

To overcome the limitations of single-method retrieval, the system combines vector similarity with graph-based expansion. Vector retrieval returns semantically similar chunks, while graph retrieval identifies structurally related evidence through entity and adjacency links.

Because the two channels produce scores in different spaces, both are normalized before fusion:

$$S_v(c) = \frac{1}{1 + \max(d_{L2}(c), 0)} \quad (1)$$

$$S_g(c) = \frac{\text{entity_hits}(c)}{\max_j(\text{entity_hits}(j))} \quad (2)$$

The final hybrid score is computed as

$$S_{\text{hybrid}}(c) = w_v S_v(c) + w_g S_g(c) \quad (3)$$

where $w_v = 0.75$ and $w_g = 0.25$. The weights were empirically selected to prioritize semantic relevance while incorporating structural signals. A cross-encoder reranker is applied after fusion to refine final context selection.

3.7 Grounded Answer Generation

Unconstrained generation can introduce unsupported claims even with correct retrieval, making grounding essential for hallucination reduction. The final answer is generated using a language model conditioned strictly on retrieved context. Generation is performed using a GPT-based model with a structured prompt that enforces evidence usage and citation formatting.

The prompt enforces citation and abstention constraints: every statement must be supported by retrieved evidence, and if sufficient support is absent, the system returns the fixed response *Not found in documents*. This ensures that the model does not generate information beyond the provided evidence.

3.8 Logging and Evaluation Integration

Structured logging is required to enable reproducible evaluation and fair comparison between retrieval modes. The system records retrieved candidates, final context, generated responses, citations, and retrieval mode for each query.

These logs support downstream evaluation of retrieval quality, grounding accuracy, and hallucination behavior. By preserving outputs from both vector and hybrid pipelines, the system enables controlled analysis under a shared evaluation framework.

IV. EXPERIMENTS AND RESULTS

The evaluation is structured to isolate distinct failure modes in the RAG pipeline and measure the effect of hybrid retrieval on evidence grounding and hallucination. Two retrieval configurations are compared: a vector-only baseline and a hybrid vector-graph system. Section IV-A describes the experimental setup, Section IV-B defines the evaluation metrics, Section IV-C presents the vector-only baseline results, Section IV-D presents hybrid retrieval results, and Section IV-E provides a comparative analysis across both configurations.

4.1 Experimental Setup

The evaluation was conducted over three documents of different types and formats: a two-column IEEE-style PDF, a structured DOCX proposal, and a short assignment PDF. This selection tests the extraction and chunking pipeline across plain text, tables, and structured layouts, ensuring that results are not specific to a single document type.

A total of 102 queries were manually created, covering six categories: single-chunk factual queries (40), multi-chunk queries (18), abstention queries (19, 18.6% of total), cross-document queries (10), entity-specific queries (9), and multi-hop queries (6). Unanswerable queries are included specifically to measure whether the system correctly abstains rather than hallucinating an answer. Both retrieval modes were evaluated against the same gold dataset and annotation set to ensure a controlled and fair comparison. The gold dataset uses source-level annotations mapping each query to one or more of the three source documents, enabling source-level recall and citation precision computation without requiring manual chunk-level tagging.

4.2 Evaluation Metrics

Seven metrics were selected to isolate distinct failure modes across retrieval, grounding, and generation stages. Separating these failure modes is necessary to determine whether any improvement from hybrid retrieval comes from better evidence coverage or from another factor.

Recall@10 measures whether chunks from the correct source document appear within the top k retrieved results. It isolates pure retrieval failure, independent of generation behavior.

Citation Precision measures the fraction of cited source documents that are genuinely relevant to the query according to the gold annotation. It isolates grounding failure: a model that cites irrelevant sources inflates unsupported responses even when retrieval succeeds.

Unsupported Claim Rate measures the fraction of generated responses that contain claims not supported by any cited evidence. It directly quantifies hallucination at the response level.

Abstention Accuracy measures whether the system correctly returns *Not found in documents.* for unanswerable queries. It evaluates hallucination control when evidence is absent.

Answer F1 measures token-level overlap between the generated answer and the gold reference string. It captures end-to-end response quality beyond retrieval and grounding alone.

Graph Contribution Rate measures the fraction of final context chunks that originated from graph-based retrieval rather than vector retrieval alone. It quantifies the additive value of the knowledge graph layer and applies only to hybrid mode.

Multihop Recall@10 measures source-level recall restricted to queries annotated as multi-hop, where the correct answer requires evidence from more than one document or chunk.

4.3 Vector-Only Baseline Results

The vector-mode evaluation was executed against all 102 queries using $k=10$ retrieved chunks. Table 2 presents the full results, which serve as the baseline for comparison against the hybrid configuration.

Table 2. Vector-Mode Baseline Evaluation Results ($k=10$)

Metric	Score	Query Count
Recall@10	95.3%	102
Abstention Accuracy	87.3%	102
Citation Precision	74.0%	102
Unsupported Claim Rate	12.7%	102
Answer F1	48.8%	83
Answer Exact Match	0.0%	83
Graph Contribution	N/A	Hybrid only
Multihop Recall@10	74.1%	9

The vector baseline results presented in Table 2 demonstrate strong overall performance. Recall@10 at 95.3% confirms the retriever locates correct source documents for nearly all queries, with the remaining gap attributed to cross-document queries where evidence spans multiple sources. Citation Precision at 74.0% shows that the majority of cited sources are relevant, with the remaining 26% attributable to over-retrieval of tangentially related chunks. Abstention Accuracy at 87.3% demonstrates strong hallucination control across 19 unanswerable queries; the remaining gap reflects cases where the model drew on semantically similar but incorrect context. Unsupported Claim Rate at 12.7% confirms that the grounded generation pipeline with strict citation enforcement effectively limits hallucination. Answer F1 at 48.8% is reasonable given the diversity of query types and the generative nature of the responses; Exact Match at 0.0% is expected for any open-ended generative model. Multihop Recall@10 at 74.1% over 9 multi-hop queries indicates that the vector retriever handles cross-document evidence reasonably well even without graph assistance. These results establish the baseline that hybrid retrieval is evaluated against in Section IV-D.

4.4 Hybrid Retrieval Results

The hybrid-mode evaluation was executed against the same 102 queries using the same $k=10$ setting, with vector weight

$w_v = 0.75$ and graph weight $w_g = 0.25$. Table 3 presents the full hybrid results.

The hybrid results presented in Table 3 demonstrate that graph-enhanced retrieval maintains equivalent answer quality to the vector baseline across all standard metrics while introducing measurable graph-sourced evidence coverage.

Table 3. Hybrid Retrieval Evaluation Results ($k=10$)

Metric	Score	Query Count
Recall@10	95.8%	102
Citation Precision	74.6%	102
Abstention Accuracy	88.5%	102
Unsupported Claim Rate	12.7%	102
Answer F1	49.0%	83
Answer Exact Match	0.0%	83
Graph Contribution	31.4%	102
Multihop Recall@10	74.9%	9

The most significant finding is the Graph Contribution Rate of 31.4%, confirming that knowledge graph retrieval actively supplements vector retrieval in nearly one-third of all queries. This rate indicates that the graph layer is not redundant: it surfaces entity-linked and structurally related chunks that the vector index does not rank highly on its own, particularly for entity-specific and cross-document queries where named entities and relational structure are more discriminative than semantic similarity alone. Beyond graph coverage, hybrid retrieval also outperforms the vector baseline across five of six comparable metrics: Recall@10 (+0.5pp), Citation Precision (+0.6pp), Abstention Accuracy (+1.2pp), Answer F1 (+0.2pp), and Multihop Recall@10 (+0.8pp).

4.5 Comparative Analysis

Table 4 presents a direct metric-by-metric comparison between the vector-only and hybrid retrieval configurations.

Table 4. Vector vs Hybrid - Comparative Results ($k=10$)

Metric	Vector	Hybrid	Delta
Recall@10	95.3%	95.8%	Hybrid +0.5pp
Citation Precision	74.0%	74.6%	Hybrid +1.6pp
Abstention Accuracy	87.3%	88.5%	Hybrid +1.2pp
Unsupported Claim Rate	12.7%	12.7%	Tied 0.0pp
Answer F1	48.8%	49.0%	Hybrid +0.3pp
Graph Contribution	N/A	31.4%	hybrid only
Multihop Recall@10	74.1%	74.9%	Hybrid +0.8pppp

The comparative analysis reveals that hybrid retrieval outperforms vector-only retrieval across five of six comparable metrics while adding substantive graph-sourced evidence coverage. Recall@10 improves from 95.3% to 95.8%, Citation Precision from 74.0% to 74.6%, Abstention Accuracy from 87.3% to 88.5%, Answer F1 from 48.8% to 49.0%, and Multihop Recall@10 from 74.1% to 74.9%. Unsupported Claim Rate remains unchanged at 12.7%, confirming that the graph layer introduces no hallucination risk. The central finding is that

hybrid retrieval is both *additive and superior*: the graph layer contributes to 31.4% of all query responses while also improving answer quality across retrieval, grounding, and generation metrics. This directly addresses the research question in Section I — knowledge graph integration expands evidence coverage for entity-specific and cross-document queries where structural relationships provide retrieval signals that semantic similarity alone cannot capture [2].

V. CONCLUSIONS AND FUTURE WORK

This work investigated whether hybrid vector-graph retrieval can reduce hallucination and improve evidence grounding in document-based question answering. The results confirm that the hybrid approach successfully addresses the core limitation of pure vector retrieval: entity-specific and cross-document queries where semantically similar but factually incomplete chunks are ranked above precise evidence. By combining dense vector similarity with structural graph traversal, the system achieves a Graph Contribution Rate of 31.4% - confirming that the knowledge graph layer actively supplements retrieval rather than remaining redundant. Critically, this coverage gain is accompanied by measurable improvements across answer quality metrics: Recall@10 (95.8%), Citation Precision (74.6%), Abstention Accuracy (88.5%), and Answer F1 (49.0%) all improve over the vector baseline, while Unsupported Claim Rate remains low at 12.7%.

A second key finding is that strict grounding constraints are essential for hallucination control independent of retrieval quality. Enforcing citation requirements and abstention behavior ensures responses remain tied to retrieved evidence, as demonstrated by the system’s fine-grained abstention behavior - answering the answerable portion of a mixed query while correctly refusing the unanswerable portion. The weighted hybrid scoring function ($w_v = 0.75$, $w_g = 0.25$) with normalized score spaces proved effective, allowing semantic relevance to dominate while graph signals provide meaningful boosts for entity-linked queries.

For future work, multi-hop graph traversal would improve performance on complex relational queries by following entity relationships at greater depth. Dynamic weight tuning could improve adaptability across query types. Replacing the API-based entity extractor with a local model would reduce cost and enable offline deployment. Claim-level NLI verification would provide a more precise hallucination metric beyond the current citation-based proxy. Evaluation on larger multi-domain datasets such as HotpotQA would enable statistically stronger validation of the hybrid approach.

REFERENCES

- [1] K. Sawarkar, A. Mangal and S. R. Solanki, “Blended RAG: Improving RAG Accuracy with Semantic Search and Hybrid Query-Based Retrievers,” *2024 IEEE 7th Int. Conf. on Multimedia Information Processing and Retrieval (MIPR)*, 2024, pp. 155–161.
- [2] B. Sarmah et al., “HybridRAG: Integrating Knowledge Graphs and Vector Retrieval Augmented Generation for Efficient Information Extraction,” *arXiv:2408.04948*, August 2024.

- [3] R. Koç, M. K. Gürkan and F. T. Yarman Vural, “ReRag: A New Architecture for Reducing Hallucination by Retrieval-Augmented Generation,” *2024 9th Int. Conf. on Computer Science and Engineering (UBMK)*, 2024, pp. 961–965.
- [4] H. Hu, C. He, X. Xie and Q. Zhang, “LRP4RAG: Detecting Hallucinations in Retrieval-Augmented Generation via Layer-Wise Relevance Propagation,” arXiv:2408.15533, August 2024.
- [5] Y. A. Yadkori et al., “Mitigating LLM Hallucinations via Conformal Abstention,” arXiv:2405.01563, April 2024.
- [6] W. T. Chan et al., “Optimizing Confidence Scoring in RAG-Based LLM Chatbots for Technical Support Services: A Prompt Engineering Approach,” *2025 IEEE 15th Symp. on Computer Applications & Industrial Electronics (ISCAIE)*, 2025, pp. 540–545.
- [7] S. Knollmeyer et al., “Document Knowledge Graph to Enhance Question Answering with Retrieval Augmented Generation,” *2024 IEEE 29th Int. Conf. on Emerging Technologies and Factory Automation (ETFA)*, 2024, pp. 1–4.